Applying Computational Intelligence to Improve Judiciary Efficiency

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***Abstract*—The present work aims to demonstrate that the use of computational intelligence in judicial provision can bring speed and legal certainty to judicial decisions. The possibility of using a technological tool based on processing in natural language and textual classification will be created and then evaluated for the simultaneous application of judicial precedents. With this, it will be sought to prove that the judicial decision can be rendered more quickly. Through the use of computational intelligence, we carried out the reading of the initial petition of processes distributed with the Special Courts of the District of Palmas-TO in Brazil in the years 2019 to 2022, focusing in the binding precedents edited by the STF and STJ. We seek the possible incidence and application of the binding precedent in the specific case and thus demonstrate that the judge could know, right after the filing of the action, that the object of the inaugural claim already had a manifestation of the STF or STJ and therefore it would be up to the judgment of injunction, and thus prove the possibility of obtaining speed and legal certainty in the jurisdictional provision with the combined use of technology with binding precedents.**

Keywords— Preliminary Dismissal of the Request; Binding Precedents; Computational Intelligence; Text Classification; Natural Language Processing

I. Introduction

In view of the significant growth of new cases that arrive in the Judiciary each year and the shortage of manpower to solve them, as already indicated by the Portal of the National Council of Justice (CNJ), known as Justice in Numbers, procedural speed, known by the constitutional legislator as the reasonable duration of the process, is compromised, and the search for alternative and technological means to undertake a satisfactory march for the resolution of disputes to achieve the effectiveness and efficiency of the jurisdictional provision.

In the specific case, simultaneous use of computational intelligence with Natural Language Processing (NLP) will be made, creating a trained algorithm with mandatory precedents, to help the magistrate to verify prematurely, at first sight of the file, the occurrence of a procedural situation that immediately authorizes the initial request to be dismissed outright, closing the dispute in a short period of time.

Identifying the processes in which the jurisprudence can be applied, it will be sought to demonstrate, in the end, that the preliminary judgment of dismissal of the request can be carried out with the aid of computational intelligence, allowing its adoption soon after the beginning of the judicial process, without the need for the actions to follow all the normal procedures, which will result in speed in the judicial provision, and it is worth noting that all identical demands will receive the same judicial decision, evidencing the legal security of the relations [7].

II. Some Basic Concepts About Brazilian Judiciary

Here, some basic concepts of the Brazilian judiciary used in this work will be presented.

A. Judgment of Preliminary Dismissal of the Request

As a tool available to the magistrate to curb those new demands whose subject has already been the subject of judicial analysis and which generated a jurisprudence. The preliminary rejection of the request implements the principles of reasonable duration of the process and legal certainty, accelerating the termination of the process by anticipating the dismissal and, at the same time, safeguarding the authority of the binding judicial precedent, thus maintaining the stability, integrity and coherence of judicial decisions

B. Binding Precedent

The system of judicial precedents aims to provide greater legal certainty, based on equal judgment for equal cases, aiming to achieve the effectiveness of justice. It is noticed that the application of mandatory precedents reduces the duration of the processes. Currently, there are more than three thousand mandatory precedents (jurisprudences), with constant changes in their content and new publications, turning the human domain over the binding system impaired, giving way to the use of technology to remedy this gap.

If the magistrate fails to apply a mandatory precedent at the beginning of the processing of the request, will cause a delay in the judicial provision, since several procedural acts be practiced only in the judgment phase, noting that it could have been judged outright as unfounded right at the beginning.

III. Computational Intelligenge in the Judgement of Preliminary Dismissal of the Request

Taking advantage of Deep Learning (DL), this work enables the automation of textual analyzes of legal processes, which in this case are the initial processes, to identify THEMES 864 and 1177, both of the STF and THEMES 986 and 1118, of the STJ, which are the jurisprudence in question.

2023 Fifth International Conference on Transdisciplinary AI (TransAI) | 979-8-3503-5801-8/23/$31.00 ©2023 IEEE | DOI: 10.1109/TRANSAI60598.2023.00027

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In this work, we focus on DL models known as LSTM (Long Short-Term Memory) and BERT (Bidirectional Encoder Representations from Transformers) [2], [4], [5].

LSTM and BERT are Deep Learning models that have proven to be very effective in a variety of NLP tasks, including text classification [9], which is the main purpose of using them in this work. Here, the text classification is done in the following way: given a text (legal process), the deep learning model gives what is the class (THEME) for it.

Now the steps of applying computational intelligence to the problem in question will be presented.

A. Dataset Generation

Initially, we have 719 processes of initial petitions in PDF files extracted from the electronic process system (Eproc/TJTO). Initially, they were manually labeled as having one of the THEMES. After that, they were read and a CSV file was generated, where the first column is the THEME it has and the second column has all the text of the respective PDF.

B. Text Pre-Processing

This step works with the text referring to each initial request of each line of data, transforming it so that it can be worked on by the DL technique, using the NLP libraries Spacy (https://spacy.io/) and NLTK (https://www.nltk.org/), which are the main existing ones [6].

The techniques used in the sequence are:

- First tokenize all text in separated words (tokens).
- Turn all tokens to lowercase, followed by removal of stopwords (mainly articles and pronouns) and all punctuation, numbers and special characters.
- Lemmatize all the tokens. Lemmatization is the process of grouping together the inflected forms of a word so they can be analyzed as a single item, identified by the word's lemma.
- Application of word embedding: this technique aims to represent a word by means of a numerical vector of low dimensionality. The technique used was BERT in Portuguese, known as BERTimbau developed by Neuralmind (https://neuralmind.ai/) [8].

C. Splitting the Data Set for Training and Testing

In this step, the data set is divided into two parts with percentage split of 90% for training and 10% for testing. It is also important to maintain the balance of each class. For this, the SMOTE technique (Synthetic Minority Over-sampling Technique) is used [1]. SMOTE is an oversampling technique used in machine learning to handle imbalanced datasets [3].

D. Deep Learning LSTM and BERT Training

The training phase is where the techniques are learned. The data are presented to the network for a number of times (epochs) and it is expected that, in the end, a certain desired learning level has been reached, that is, that a minimum value of a selected learning evaluation measure has been obtained.

E. Training Evaluation

To evaluate the training performance, evaluation metrics commonly used in classification problems were used, such as confusion matrix, accuracy, and F1-Score. A confusion matrix is an artifact used to evaluate the performance of models intended for performing classification in machine learning.

If training has reached a value greater than or equal to the established performance metric (e.g. accuracy or F1-Score), we move on to testing. Otherwise, it returns to the previous training stage in order to improve their learning/training (change the network architecture and/or its hyperparameters).

F. Deep Learning LSTM and BERT Test

Here, LSTM and BERT are evaluated with different data of the learning phase to verify the performance for unknown data in order to know if the learning process was effective. With LSTM and BERT presenting performance greater than or equal to a certain value, in the process of correct classification of cases, the next phase is the application in production. Otherwise, with low performance, we return to step of network training in order to improve network training and, consequently, performance in the test phase.

G. Generated Model

Once the model is ready, both the LSTM and the BERT are applied to any new lawsuit filed, with the magistrate seeking to find similarity between the new request and one of the THEMES, with the aim of discovering whether the mandatory precedent (THEME) has applicability to the case and thus proceeding with the injunction of dismissal, gaining speed in the provision of jurisdiction, with legal security.

IV. Computational Experiments and Results Analysis

Here, the computational results of the DL used and their effectiveness in correctly estimating whether or not an initial petition has any of the specified THEMES is presented.

A. Configuration of the Performed Computational Experiment

Fig. 1 and Fig. 2 show the Deep Learning LSTM and BERT architectures used in this work.

```
Layer (type)                 Output Shape              Param #
=================================================================
embedding (Embedding)        (None, 100, 100)          5000000

spatial_dropout1d (SpatialD  (None, 100, 100)          0
ropout1D)

lstm (LSTM)                  (None, 100)               80400

dense (Dense)                (None, 5)                 505

=================================================================
Total params: 5,080,905
Trainable params: 5,080,905
Non-trainable params: 0
```

Figure 1: Deep Learning LSTM architecture used.

```
Layer (type)                 Output Shape              Param #
=================================================================
input_ids (InputLayer)       [(None, 30)]              0

bert (TFBertMainLayer)       TFBaseModelOutputWithPoo  108923136
                             lingAndCrossAttentions(l
                             ast_hidden_state=(None,
                             30, 768),
                              pooler_output=(None, 76
                             8),
                              past_key_values=None, h
                             idden_states=None, atten
                             tions=None, cross_attent
                             ions=None)

pooled_output (Dropout)      (None, 768)               0

tipodecisao02 (Dense)        (None, 25)                19225

tipodecisao01 (Dropout)      (None, 25)                0

tipodecisao (Dense)          (None, 5)                 130

=================================================================
Total params: 108,942,491
Trainable params: 108,942,491
Non-trainable params: 0
```

Figure 2: Deep Learning BERT architecture used.

All experiments were performed on a machine with Windows 10 operating system, Intel i7-11800H 2.30Ghz

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processor, 16 Gb of RAM and RTX3070 GPU.

B. *Results of the Experiment Performed with the Deep Learning LSTM and BERT*

For a computational effort (runtime) analysis, LSTM took 127.10 seconds in pre-processing, 88.54 seconds in training and 0.62 seconds in the testing for the presented dataset.

LSTM with the configuration presented in Fig. 1 presented an accuracy of 97% in the training phase and an accuracy of 90% in the testing phase. To corroborate this information, Fig. 3 presents the confusion matrix for the test data.

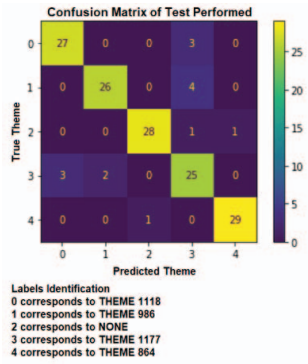


Figure 3: Confusion matrix on the LSTM test data.

The confusion matrix in Fig. 3 shows that for the 5 classes with 150 examples, only 15 are misclassified (about 10%).

From the data in Fig. 3, we generate the classification report in Fig. 4, with the most used metrics for evaluating the performance of computational intelligence classification techniques, which are precision, recall and F1-Score.

From Fig. 4, in the first block of values, the performance measures for each of the classes are presented separately. In the second block of values, the average measures of all classes are presented. All metrics vary between 0 and 1 and the closer to 1 the better. The average value of 0.90 for all classes, showing a good performance of the proposed tool.

For a computational effort analysis, BERT took 198.28 seconds in pre-processing, 652.30 seconds in training, and 0.92 seconds in testing.

For the DL BERT with the configuration of Fig. 2, presents an accuracy of 99.70% in the training phase and an accuracy of 99% in the testing phase. Confirming this, Fig. 5 presents the confusion matrix of DL BERT for the test data.

	precision	recall	f1-score	support
0	0.90	0.90	0.90	30
1	0.87	0.93	0.90	28
2	0.93	0.97	0.95	29
3	0.83	0.76	0.79	33
4	0.97	0.97	0.97	30
micro avg	0.90	0.90	0.90	150
macro avg	0.90	0.90	0.90	150
weighted avg	0.90	0.90	0.90	150
samples avg	0.90	0.90	0.90	150

Figure 4: Measures of LSTM's test classification report.

The confusion matrix in Fig. 5 shows that for the 5 classes with 150 examples, only 2 are misclassified (about 1%), leading to a classification accuracy of 99% in the test data. Here, the following errors occurred: 2 instances of class 2 (NONE) were misclassified as class 4 (THEME 864).

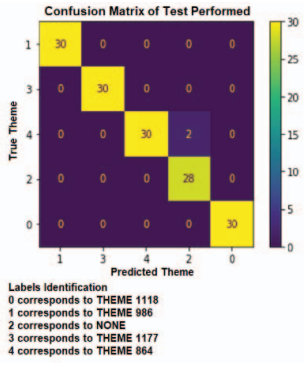


Figure 5: Confusion matrix on the BERT test data.

From the data in Fig. 5, the classification report in Fig. 6 is generated with the most used metrics for evaluating the performance of computational intelligence classification techniques, which are precision, recall and F1-Score.

In Fig. 6, in the first block of values, the performance measures for each of the classes are presented separately. In the second block of values, the average measures of all classes are presented. All these metrics vary between 0 and 1 and the closer to 1 the better. In the case at hand, the average value of 0.99 for all classes, shows a good performance for BERT.

	precision	recall	f1-score	support
0	1.00	1.00	1.00	30
1	1.00	1.00	1.00	30
2	0.93	1.00	0.97	28
3	1.00	1.00	1.00	30
4	1.00	0.94	0.97	32
accuracy			0.99	150
macro avg	0.99	0.99	0.99	150
weighted avg	0.99	0.99	0.99	150

Figure 6: Measures of BERT's test classification report.

C. *Analysis of Obtained Results*

Observing the results of the Deep Learning LSTM and BERT presented, the following notes are obtained:

- DL BERT presents higher computational time than DL LSTM, in all phases (training, testing and prediction/estimation), mainly in the training phase.
- In the prediction/estimation phase, the computational times are 15ms to LSTM and 98ms to BERT.
- DL BERT performed better than DL LSTM in all evaluation metrics, in particular, with an average performance of 99% for BERT against 90% for LSTM in testing phase, characterizing superior performance.

From the notes, it appears that DL BERT has greater computational time than DL LSTM in all phases (training, testing and prediction/estimation), however, in the prediction/estimation phase, at which time the occurrence or not of any legal precedent is identified in the initial petition, there is a negligible difference between the networks (98ms for BERT and 15ms for LSTM = 83ms) for the end user. Nevertheless, the performance of BERT (99%) performed 9% better than LSTM (90%) to determine whether or not there is any binding precedent and, if so, will indicate which one.

Therefore, we concluded that BERT is superior than LSTM for the problem in question and is appropriate for estimating THEMES from the presented exordials.

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Subsequently, an additional evaluation was carried out with 100 processes different from those used in the training and testing phases. In this case, the accuracy obtained was 99%, demonstrating the high success rate of the proposed tool. Noting that the processing time of each individual process takes an average of 3.57 seconds, showing the high efficiency of the proposed tool in the phase of analysis and identification of similarity of some THEME with the initial petition.

The proposed technological tool, given an initial petition, first goes through a text pre-processing phase. Afterwards, these pre-processed text data are passed on to the technological tool for DL used to perform the prediction/estimation. At the end, it is presented whether NONE or which THEME is applied to the analyzed process. After, the section referring to the lawyer's request, requirements or final requirements in the process, as well as the link to the jurisprudence in the superior courts (STF or STJ) with all the information corresponding to the precedent are also presented, to facilitate the analysis of the answer given by the technological tool to the judge, remembering that the same seeks to make the judgment faster and more efficient and not to replace the magistrate in the practice of his acts.

V. Conclusions and Future Works

It was demonstrated that the use of computational intelligence will help the judge to discover the incidence of some precedent right at the beginning of the process, well after the filing of the initial petition. The proposed tool seeks to help the magistrate in the task of finding the incidence of some precedent in the demand that is under his analysis in a quick and efficient way. It provides this in less than 1 second on the machine used. Nevertheless, the proposed tool (based on DL BERT) presented a 99% accuracy margin on unseen data, presenting an excellent performance.

Here we do not want to create a technological tool that will render judicial decisions in place of the judge, but a form of technological assistance in the search for a quick and safe judicial provision. We teach the machine to analyze the body of the initial petition and compare it with the text of the precedents so that the magistrate does not fail to apply a decision that has already been pacified to the case or that gives a conflicting result for a similar situation of law.

The tool only helps the magistrate in finding the similarity between the initial request and the existence of some precedent of mandatory observance, not replacing the judge in the analysis of the dispute, much less imposing the practice of some decision-making act on the demand. It will always be up to the magistrate to analyze the occurrence of any binding precedent to the concrete case and thus adopt, if applicable, the preliminary injunction judgment of the request, making the judicial provision safer, quicker and more productive.

As previously noted, the wide range of precedents makes difficult for the judge to master its content in general, even more so when it comes to a new member in the career or when there is a change in competence regarding the matter in the court where the magistrate operates, situations in which the aid of the technological tool proves to be essential for the good and speedy exercise of the jurisdiction.

In this sense, the development of a multidisciplinary tool that combines law, artificial intelligence and systems design certainly contributes to other areas of the Judiciary, and Public Administration as a whole, to improve their working methods, making them more agile and efficient.

As future work, it is intended to effectively implement the technological tool developed in this work to the EProc/TJTO system both in the first and second degrees of jurisdiction, making the necessary adjustments so that all the thousands of binding precedents are included in the technological tool, so that it can effectively assist the magistrates in the distribution of a faster jurisdiction, being important to make the Court of Justice of the State of Tocantins aware of the need to implement and operate an artificial intelligence laboratory for the development and creation of other digital instruments.

This insertion and constant updating of precedents in the technological tool must be carried out and monitored by the Precedent Management Center (NUGEP), which is the body responsible for cataloging and disseminating judicial precedents formed by the judgment of repetitive demands and incidents of assumption of jurisdiction, as well as for the standardization for the application of these precedents in pending and future judicial proceedings.

Other future work, is the proposal to divide the binding precedents (jurisprudences) according to the subjects of the proceedings, in each court. With this, an attempt will be made to adapt the technological tool, training the DL BERT, only with the binding precedents corresponding to the matter of competence of that court, thus seeking an improved performance, given that the number of binding precedents applicable to a given court is much smaller than the total.

Nevertheless, we also plan to study trends and biases in the results presented by the proposed technique and find methods that solve this problem.

Acknowledgment

All authors thanks ESMAT/TJTO, METROTINS and FAPTO for the partnership and financial support.

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